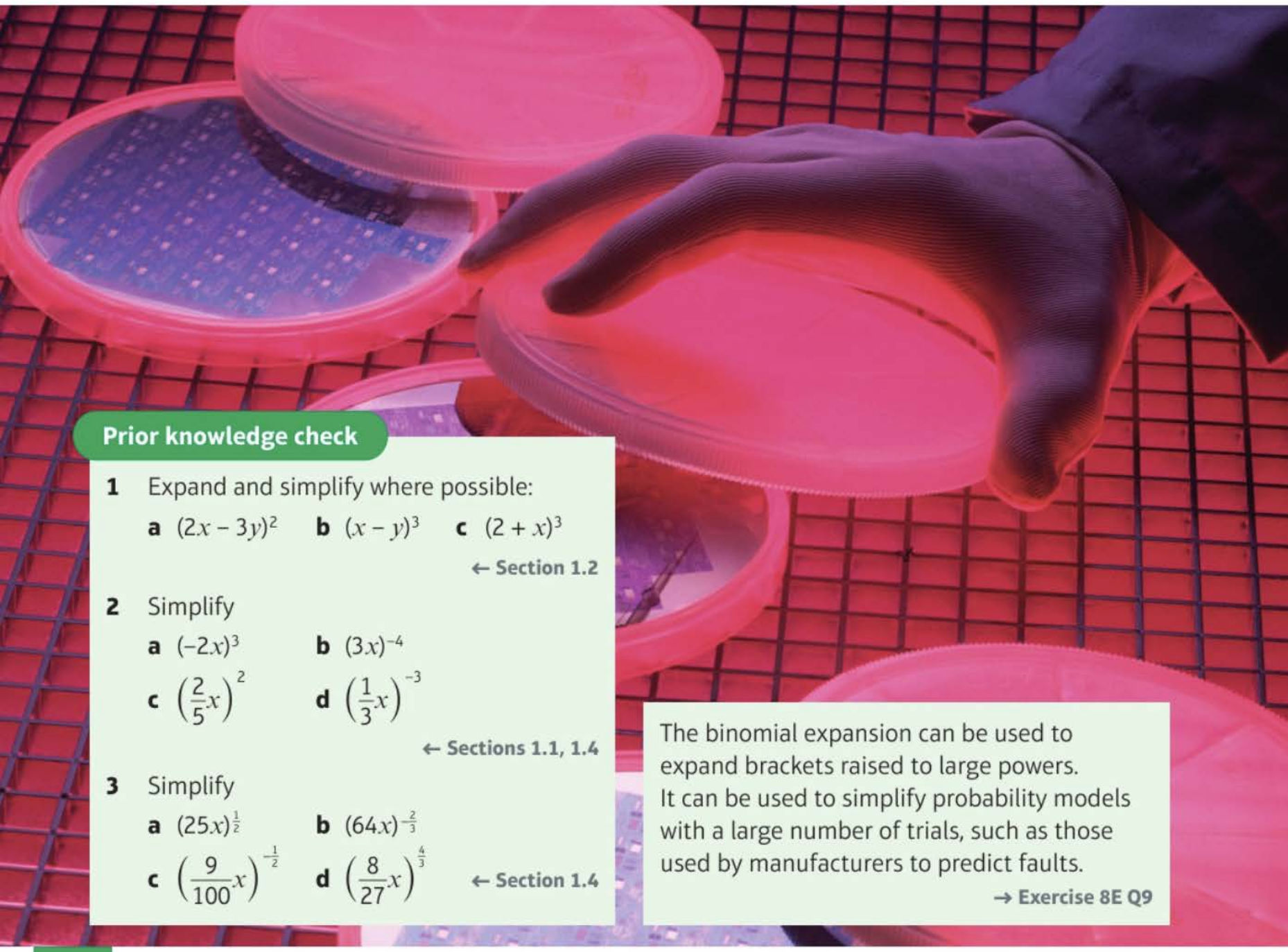
8 The binomial expansion

Objectives

After completing this chapter you should be able to:

- Use Pascal's triangle to identify binomial coefficients and use them to expand simple binomial expressions → pages 159–161
- Use combinations and factorial notation → pages 161–163
- Use the binomial expansion to expand brackets → pages 163–165
- Find individual coefficients in a binomial expansion → pages 165–167
- Make approximations using the binomial expansion → pages 167–169



Prior knowledge check

1 Expand and simplify where possible:

a $(2x - 3y)^2$ **b** $(x - y)^3$ **c** $(2 + x)^3$

← Section 1.2

2 Simplify

a $(-2x)^3$ **b** $(3x)^{-4}$

c $\left(\frac{2}{5}x\right)^2$ **d** $\left(\frac{1}{3}x\right)^{-3}$

← Sections 1.1, 1.4

3 Simplify

a $(25x)^{\frac{1}{2}}$ **b** $(64x)^{-\frac{2}{3}}$

c $\left(\frac{9}{100}x\right)^{-\frac{1}{2}}$ **d** $\left(\frac{8}{27}x\right)^{\frac{4}{3}}$ ← Section 1.4

The binomial expansion can be used to expand brackets raised to large powers. It can be used to simplify probability models with a large number of trials, such as those used by manufacturers to predict faults.

→ Exercise 8E Q9

8.1 Pascal's triangle

You can use **Pascal's triangle** to quickly expand expressions such as $(x + 2y)^3$.

Consider the expansions of $(a + b)^n$ for $n = 0, 1, 2, 3$ and 4:

$(a + b)^0 = 1$

$(a + b)^1 = 1a + 1b$

$(a + b)^2 = 1a^2 + 2ab + 1b^2$

$(a + b)^3 = 1a^3 + 3a^2b + 3ab^2 + 1b^3$

$(a + b)^4 = 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4$

Each coefficient is the sum of the two coefficients immediately above it.

Every term in the expansion of $(a + b)^n$ has total index n:
In the $6a^2b^2$ term the total index is $2 + 2 = 4$.
In the $4ab^3$ term the total index is $1 + 3 = 4$.

The coefficients in the expansions form a pattern that is known as Pascal's triangle.

- **Pascal's triangle is formed by adding adjacent pairs of numbers to find the numbers on the next row.**

Here are the first 7 rows of Pascal's triangle:

1
1 1
1 + 2 1
1 3 + 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1

The third row of Pascal's triangle gives the coefficients in the expansion of $(a + b)^2$.

- **The $(n + 1)$th row of Pascal's triangle gives the coefficients in the expansion of $(a + b)^n$.**

Example 1

Use Pascal's triangle to find the expansions of:

a $(x + 2y)^3$ **b** $(2x - 5)^4$

a $(x + 2y)^3$

The coefficients are 1, 3, 3, 1 so:

$(x + 2y)^3 = 1x^3 + 3x^2(2y) + 3x(2y)^2 + 1(2y)^3$

$= x^3 + 6x^2y + 12xy^2 + 8y^3$

Index = 3 so look at the 4th row of Pascal's triangle to find the coefficients.

This is the expansion of $(a + b)^3$ with $a = x$ and $b = 2y$. Use brackets to make sure you don't make a mistake. $(2y)^2 = 4y^2$.

b $(2x - 5)^4$

The coefficients are 1, 4, 6, 4, 1 so:

$(2x - 5)^4 = 1(2x)^4 + 4(2x)^3(-5)^1 + 6(2x)^2(-5)^2 + 4(2x)^1(-5)^3 + 1(-5)^4$

$= 16x^4 - 160x^3 + 600x^2 - 1000x + 625$

Index = 4 so look at the 5th row of Pascal's triangle.

This is the expansion of $(a + b)^4$ with $a = 2x$ and $b = -5$.

Be careful with the negative numbers.

Example 2

The coefficient of x^2 in the expansion of $(2 - cx)^3$ is 294. Find the possible value(s) of the constant c.

Problem-solving

If there is an unknown in the original expression, you might be able to form an equation involving that unknown.

The coefficients are 1, 3, 3, 1:

The term in x^2 is $3 \times 2(-cx)^2 = 6c^2x^2$

So $6c^2 = 294$

$c^2 = 49$

$c = \pm 7$

Index = 3 so use the 4th row of Pascal's triangle.

From the expansion of $(a + b)^3$ the x^2 term is $3ab^2$ where $a = 2$ and $b = -cx$.

Form and solve an equation in c.

Exercise 8A

1 State the row of Pascal's triangle that would give the coefficients of each expansion:

a $(x + y)^3$ **b** $(3x - 7)^{15}$ **c** $\left(2x + \frac{1}{2}\right)^n$ **d** $(y - 2x)^{n+4}$

2 Write down the expansion of:

a $(x + y)^4$ **b** $(p + q)^5$ **c** $(a - b)^3$ **d** $(x + 4)^3$

e $(2x - 3)^4$ **f** $(a + 2)^5$ **g** $(3x - 4)^4$ **h** $(2x - 3y)^4$

3 Find the coefficient of x^3 in the expansion of:

a $(4 + x)^4$ **b** $(1 - x)^5$ **c** $(3 + 2x)^3$ **d** $(4 + 2x)^5$

e $(2 + x)^6$ **f** $\left(4 - \frac{1}{2}x\right)^4$ **g** $(x + 2)^5$ **h** $(3 - 2x)^4$

(P) **4** Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

Problem-solving

Expand $(1 + 2x)^3$, then multiply each term by 1 and by $3x$.

(P) **5** Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

(P) **6** The coefficient of x^2 in the expansion of $(2 + ax)^3$ is 54. Find the possible values of the constant a.

P **7** The coefficient of x^3 in the expansion of $(2 - x)(3 + bx)^3$ is 45. Find possible values of the constant b.

P **8** Work out the coefficient of x^2 in the expansion of $(p - 2x)^3$. Give your answer in terms of p.

P **9** After 5 years, the value of an investment of £500 at an interest rate of $X\%$ per annum is given by:

$$500\left(1 + \frac{X}{100}\right)^5$$

Find an approximation for this expression in the form $A + BX + CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

Challenge

Find the constant term in the expansion of $\left(x^2 - \frac{1}{2x}\right)^3$.

8.2 Factorial notation

You can use combinations and factorial notation to help you expand binomial expressions. For larger indices, it is quicker than using Pascal's triangle.

Using **factorial notation** $3 \times 2 \times 1 = 3!$

Notation You say 'n factorial'. By definition, $0! = 1$.

- **You can use factorial notation and your calculator to find entries in Pascal's triangle quickly.**
- **The number of ways of choosing r items from a group of n items is written as ${}^nC_r$ or $\binom{n}{r}$:**

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- **The rth entry in the nth row of Pascal's triangle is given by ${}^{n-1}C_{r-1} = \binom{n-1}{r-1}$**

Notation You can say 'n choose r' for ${}^nC_r$. It is sometimes written without superscripts and subscripts as nCr.

Example 3

Calculate:

a 5! **b** ${}^5C_2$ **c** the 6th entry in the 10th row of Pascal's triangle

a $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

b ${}^5C_2 = \frac{5!}{2!3!} = \frac{120}{12} = 10$

c ${}^9C_5 = 126$

Online Use the ${}^nC_r$ and ! functions on your calculator to answer this question.

You can calculate ${}^5C_2$ by using the ${}^nC_r$ function on your calculator.

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{5!}{2!(5-2)!}$$

In the expansion of $(a + b)^9$ this would give the term $126a^4b^5$.

The rth entry in the nth row is ${}^{n-1}C_{r-1}$.

Exercise 8B

1 Work out:

a 4! **b** 9! **c** $\frac{10!}{7!}$ **d** $\frac{15!}{13!}$

2 Without using a calculator, work out:

a $\binom{4}{2}$ **b** $\binom{6}{4}$ **c** ${}^{6}C_{3}$ **d** $\binom{5}{4}$ **e** ${}^{10}C_{8}$ **f** $\binom{9}{5}$

3 Use a calculator to work out:

a $\binom{15}{6}$ **b** ${}^{10}C_{7}$ **c** $\binom{20}{10}$ **d** $\binom{20}{17}$ **e** ${}^{14}C_{9}$ **f** ${}^{18}C_{5}$

4 Write each value a to d from Pascal's triangle using ${}^{n}C_{r}$ notation:

1
1 1
1 2 1
1 3 3 1
1 a 6 4 1
1 5 b 10 5 1
1 6 c d 15 6 1

5 Work out the 5th number on the 12th row from Pascal's triangle.

6 The 11th row of Pascal's triangle is shown below.

1 10 45 … …

a Find the next two values in the row.

b Hence find the coefficient of x^3 in the expansion of $(1 + 2x)^{10}$.

7 The 14th row of Pascal's triangle is shown below.

1 13 78 … …

a Find the next two values in the row.

b Hence find the coefficient of x^4 in the expansion of $(1 + 3x)^{13}$.

8 The probability of throwing exactly 10 heads when a fair coin is tossed 20 times is given by $\binom{20}{10}0.5^{20}$. Calculate the probability and describe the likelihood of this occurring.

(P) **9** Show that:

a ${}^{n}C_{1} = n$ **b** ${}^{n}C_{2} = \frac{n(n-1)}{2}$

(E) **10** Given that $\binom{50}{13} = \frac{50!}{13!a!}$, write down the value of a. **(1 mark)**

(E) **11** Given that $\binom{35}{p} = \frac{35!}{p!18!}$, write down the value of p. **(1 mark)**

Challenge

a Work out ${}^{10}C_3$ and ${}^{10}C_7$

b Work out ${}^{14}C_5$ and ${}^{14}C_9$

c What do you notice about your answers to parts **a** and **b**?

d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$

8.3 The binomial expansion

The binomial expansion is a rule that allows you to **expand brackets**. You can use $\binom{n}{r}$ to work out the coefficients in the binomial expansion. For example,

in the expansion of $(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$, to find the b^3 term you can choose multiples of b from 3 different brackets. You can do this in $\binom{5}{3}$ ways so the b^3 term is $\binom{5}{3}a^2b^3$.

■ **The binomial expansion is:**

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n \quad (n \in \mathbb{N})$$

where $\binom{n}{r} = {}^{n}C_r = \dfrac{n!}{r!(n-r)!}$

Notation $n \in \mathbb{N}$ means that n must be a member of the **natural numbers**. This is all the **positive integers**.

Example 4

Use the binomial theorem to find the expansion of $(3-2x)^5$.

$$(3-2x)^5 = 3^5 + \binom{5}{1}3^4(-2x) + \binom{5}{2}3^3(-2x)^2 + \binom{5}{3}3^2(-2x)^3 + \binom{5}{4}3^1(-2x)^4 + (-2x)^5$$
$$= 243 - 810x + 1080x^2 - 720x^3 + 240x^4 - 32x^5$$

There will be 6 terms.

Each term has a total index of 5.

Use $(a+b)^n$ where $a = 3$, $b = -2x$ and $n = 5$.

There are $\binom{5}{2}$ ways of choosing two '$-2x$' terms from five brackets.

Online Work out each coefficient quickly using the ${}^{n}C_r$ and power functions on your calculator.

Example 5

Find the first four terms in the binomial expansion of:

a $(1+2x)^{10}$ **b** $\left(10-\frac{1}{2}x\right)^6$

a $(1 + 2x)^{10}$

$= 1^{10} + \binom{10}{1}1^9(2x) + \binom{10}{2}1^8(2x)^2$

$+ \binom{10}{3}1^7(2x)^3 + \ldots$

$= 1 + 20x + 180x^2 + 960x^3 + \ldots$

b $(10 - \frac{1}{2}x)^6$

$= 10^6 + \binom{6}{1}10^5(-\frac{1}{2}x) + \binom{6}{2}10^4(-\frac{1}{2}x)^2$

$+ \binom{6}{3}10^3(-\frac{1}{2}x)^3 + \ldots$

$= 1000000 - 300000x + 37500x^2$

$- 2500x^3 + \ldots$

Write each coefficient in its simplest form.

Notation This is sometimes called the expansion in **ascending powers of** x.

Exercise 8C

1 Write down the expansion of the following:

a $(1 + x)^4$ **b** $(3 + x)^4$ **c** $(4 - x)^4$ **d** $(x + 2)^6$ **e** $(1 + 2x)^4$ **f** $(1 - \frac{1}{2}x)^4$

2 Use the binomial theorem to find the first four terms in the expansion of:

a $(1 + x)^{10}$ **b** $(1 - 2x)^5$ **c** $(1 + 3x)^6$ **d** $(2 - x)^8$ **e** $(2 - \frac{1}{2}x)^{10}$ **f** $(3 - x)^7$

3 Use the binomial theorem to find the first four terms in the expansion of:

a $(2x + y)^6$ **b** $(2x + 3y)^5$ **c** $(p - q)^8$ **d** $(3x - y)^6$ **e** $(x + 2y)^8$ **f** $(2x - 3y)^9$

4 Use the binomial expansion to find the first four terms, in ascending powers of x, of:

a $(1 + x)^8$ **b** $(1 - 2x)^6$ **c** $\left(1 + \frac{x}{2}\right)^{10}$

d $(1 - 3x)^5$ **e** $(2 + x)^7$ **f** $(3 - 2x)^3$

g $(2 - 3x)^6$ **h** $(4 + x)^4$ **i** $(2 + 5x)^7$

Hint Your answers should be in the form $a + bx + cx^2 + dx^3$ where a, b, c and d are numbers.

E **5** Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2 - x)^6$ and simplify each term. **(4 marks)**

E **6** Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3 - 2x)^5$ giving each term in its simplest form. **(4 marks)**

E/P **7** Find the binomial expansion of $\left(x + \frac{1}{x}\right)^5$ giving each term in its simplest form. **(4 marks)**

Challenge

a Show that $(a + b)^4 - (a - b)^4 = 8ab(a^2 + b^2)$.

b Given that $82896 = 17^4 - 5^4$, write 82896 as a product of its prime factors.

8.4 Solving binomial problems

You can use the general term of the binomial expansion to find individual coefficients in a binomial expansion.

- **In the expansion of $(a + b)^n$ the general term is given by $\binom{n}{r}a^{n-r}b^r$.**

Example 6

a Find the coefficient of x^4 in the binomial expansion of $(2 + 3x)^{10}$.

b Find the coefficient of x^3 in the binomial expansion of $(2 + x)(3 - 2x)^7$.

a x^4 term $= \binom{10}{4}2^6(3x)^4$

$= 210 \times 64 \times 81x^4$

$= 1088640x^4$

The coefficient of x^4 in the binomial expansion of $(2 + 3x)^{10}$ is 1088640.

Use the general term. The power is 10, so $n = 10$, and you need to find the x^4 term so $r = 4$.

There are $\binom{10}{4}$ ways of choosing 4 '$3x$' terms from 10 brackets.

b $(3 - 2x)^7$

$= 3^7 + \binom{7}{1}3^6(-2x) + \binom{7}{2}3^5(-2x)^2$

$+ \binom{7}{3}3^4(-2x)^3 + \ldots$

$= 2187 - 10206x + 20412x^2$

$- 22680x^3 + \ldots$

First find the first four terms of the binomial expansion of $(3 - 2x)^7$.

$(2 + x)(2187 - 10206x + 20412x^2$

$- 22680x^3 + \ldots)$

Now expand the brackets $(2 + x)(3 - 2x)^7$.

x^3 term $= 2 \times (-22680x)^3 + x \times 20412x^2$

$= -24948x^3$

The coefficient of x^3 in the binomial expansion of $(2 + x)(3 - 2x)^7$ is -24948.

There are two ways of making the x^3 term: (constant term $\times$ x^3 term) and (x term $\times$ x^2 term).

Example 7

$g(x) = (1 + kx)^{10}$, where k is a constant.

Given that the coefficient of x^3 in the binomial expansion of $g(x)$ is 15, find the value of k.

x^3 term $= \binom{10}{3}1^7(kx)^3 = 15x^3$

$a = 1$, $b = kx$, $n = 10$ and $r = 3$.

$120k^3x^3 = 15x^3$

$k^3x^3 = \frac{15}{120}x^3$

$k^3x^3 = \frac{1}{8}x^3$

$k = \frac{1}{2}$

$k^3 = \frac{1}{8}$, $k = \sqrt[3]{\frac{1}{8}}$

Example 8

a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^8$, where q is a non-zero constant.

b Given that, in the expansion of $(1 + qx)^8$, the coefficient of x is $-r$ and the coefficient of x^2 is $7r$, find the value of q and the value of r.

a $(1 + qx)^8$

$= 1^8 + \binom{8}{1}1^7(qx)^1 + \binom{8}{2}1^6(qx)^2 + \ldots$

$= 1 + 8qx + 28q^2x^2 + \ldots$

b $8q = -r$ and $28q^2 = 7r$

$8q = -4q^2$

$4q^2 + 8q = 0$

$4q(q + 2) = 0$

$q = -2, r = 16$

Problem-solving

There are two unknowns in this expression. Your expansion will be in terms of q and x.

Using $28q^2 = 7r$, $r = 4q^2$ and $-r = -4q^2$.

q is non-zero so $q = -2$.

Exercise 8D

1 Find the coefficient of x^3 in the binomial expansion of:

a $(3 + x)^5$	**b** $(1 + 2x)^5$	**c** $(1 - x)^6$	**d** $(3x + 2)^5$
e $(1 + x)^{10}$	**f** $(3 - 2x)^6$	**g** $(1 + x)^{20}$	**h** $(4 - 3x)^7$
i $\left(1 - \frac{1}{2}x\right)^6$	**j** $\left(3 + \frac{1}{2}x\right)^7$	**k** $\left(2 - \frac{1}{2}x\right)^8$	**l** $\left(5 + \frac{1}{4}x\right)^5$

(P) **2** The coefficient of x^2 in the expansion of $(2 + ax)^6$ is 60. Find two possible values of the constant a.

Problem-solving

$a = 2$, $b = ax$, $n = 6$. Use brackets when you substitute ax.

(P) **3** The coefficient of x^3 in the expansion of $(3 + bx)^5$ is -720. Find the value of the constant b.

(P) **4** The coefficient of x^3 in the expansion of $(2 + x)(3 - ax)^4$ is 30. Find two possible values of the constant a.

(E/P) **5** When $(1 - 2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, use this information to find:

a the value of the constant p **(6 marks)**

b the coefficient of x **(1 mark)**

c the coefficient of x^3 **(2 marks)**

Problem-solving

You will need to use the definition of $\binom{n}{r}$ to find an expression for $\binom{p}{2}$.

(E/P) **6** **a** Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant. **(2 marks)**

b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p. **(4 marks)**

E/P **7** **a** Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant. **(2 marks)**

b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q. **(4 marks)**

E/P **8** **a** Find the first three terms, in ascending powers of x of the binomial expansion of $(1 + px)^{11}$, where p is a constant. **(2 marks)**

b The first 3 terms in the same expansion are 1, $77x$ and qx^2, where q is a constant. Find the value of p and the value of q. **(4 marks)**

E/P **9** **a** Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{15}$, where p is a non-zero constant. **(2 marks)**

b Given that, in the expansion of $(1 + px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q. **(4 marks)**

E/P **10** In the binomial expansion of $(1 + x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$. **(4 marks)**

Challenge

Find the coefficient of x^4 in the binomial expansion of: **a** $(3 - 2x^2)^9$ **b** $\left(\frac{5}{x} + x^2\right)^8$

8.5 Binomial estimation

In engineering and science, it is often useful to find simple **approximations** for complicated functions. If the value of x is less than 1, then x^n gets smaller as n gets larger. If x is small you can sometimes **ignore large powers** of x to approximate a function or estimate a value.

Example 9

a Find the first four terms of the binomial expansion, in ascending powers of x, of $\left(1 - \frac{x}{4}\right)^{10}$.

b Use your expansion to estimate the value of 0.975^{10}, giving your answer to 4 decimal places.

a $\left(1 - \frac{x}{4}\right)^{10}$

$= 1^{10} + \binom{10}{1}1^9\left(-\frac{x}{4}\right) + \binom{10}{2}1^8\left(-\frac{x}{4}\right)^2$

$+ \binom{10}{3}1^7\left(-\frac{x}{4}\right)^3 + \ldots$

$= 1 - 2.5x + 2.8125x^2 - 1.875x^3 + \ldots$

b We want $\left(1-\frac{x}{4}\right)=0.975$

$$\frac{x}{4}=0.025$$

$$x=0.1$$

Substitute $x = 0.1$ into the expansion for $\left(1-\frac{x}{4}\right)^{10}$ from part **a**:

$$0.975^{10} \approx 1 - 0.25 + 0.028125 - 0.001875$$

$$= 0.77625$$

$0.975^{10} \approx 0.7763$ to 4 d.p.

Calculate the value of x.

Substitute $x = 0.1$ into your expansion.

Using a calculator, $0.975^{10} = 0.776\,329\,62$.
So approximation is correct to 4 decimal places.

Online Use GeoGebra to find the values of x for which the first four terms of this expansion give a good approximation to the value of the function.

Exercise 8E

1 a Find the first four terms of the binomial expansion, in ascending powers of x, of $\left(1-\frac{x}{10}\right)^6$.

b By substituting an appropriate value for x, find an approximate value for 0.99^6.

2 a Write down the first four terms of the binomial expansion of $\left(2+\frac{x}{5}\right)^{10}$.

b By substituting an appropriate value for x, find an approximate value for 2.1^{10}.

(P) **3** If x is so small that terms of x^3 and higher can be ignored, show that:

$$(2 + x)(1 - 3x)^5 \approx 2 - 29x + 165x^2$$

Hint Start by using the binomial expansion to expand $(1 - 3x)^5$. You can ignore terms of x^3 and higher so you only need to expand up to and including the x^2 term.

(P) **4** If x is so small that terms of x^3 and higher can be ignored, and

$$(2 - x)(3 + x)^4 \approx a + bx + cx^2$$

find the values of the constants a, b and c.

Problem-solving

Find the first 3 terms in the expansion of $(2 - x)(3 + x)^4$, compare with $a + bx + cx^2$ and write down the values of a, b and c.

5 a Write down the first four terms in the expansion of $(1 + 2x)^8$.

b By substituting an appropriate value of x (which should be stated), find an approximate value of 1.02^8.

6 $f(x) = (1 - 5x)^{30}$

a Find the first four terms, in ascending powers of x, in the binomial expansion of $f(x)$.

b Use your answer to part **a** to estimate the value of $(0.995)^{30}$, giving your answer to 6 decimal places.

c Use your calculator to evaluate 0.995^{30} and calculate the percentage error in your answer to part **b**.

(E/P) **7 a** Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form. **(4 marks)**

b Explain how you would use your expansion to give an estimate for the value of 2.98^{10}. **(1 mark)**

E **8 a** Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1 - 3x)^5$. Give each term in its simplest form. **(4 marks)**

b If x is small, so that x^2 and higher powers can be ignored, show that $(1 + x)(1 - 3x)^5 \approx 1 - 14x$. **(2 marks)**

E/P **9** A microchip company models the probability of having no faulty chips on a single production run as:

$$\text{P(no fault)} = (1 - p)^n, p < 0.001$$

where p is the probability of a single chip being faulty, and n being the total number of chips produced.

a State why the model is restricted to small values of p. **(1 mark)**

b Given that $n = 200$, find an approximate expression for P(no fault) in the form $a + bp + cp^2$. **(2 marks)**

c The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case. **(4 marks)**

Mixed exercise 8

P **1** The 16th row of Pascal's triangle is shown below.

1 15 105 … …

a Find the next two values in the row.

b Hence find the coefficient of x^3 in the expansion of $(1 + 2x)^{15}$.

E **2** Given that $\binom{45}{17} = \frac{45!}{17!a!}$, write down the value of a. **(1 mark)**

3 20 people play a game at a school fete.
The probability that exactly n people win a prize is modelled as $\binom{20}{n}p^n(1 - p)^{20-n}$, where p is the probability of any one person winning.

Calculate the probability of:

a 5 people winning when $p = \frac{1}{2}$

b nobody winning when $p = 0.7$

c 13 people winning when $p = 0.6$

Give your answers to 3 significant figures.

E/P **4** When $\left(1 - \frac{3}{2}x\right)^p$ is expanded in ascending powers of x, the coefficient of x is -24.

a Find the value of p. **(2 marks)**

b Find the coefficient of x^2 in the expansion. **(3 marks)**

c Find the coefficient of x^3 in the expansion. **(1 mark)**

E/P **5** Given that:

$$(2 - x)^{13} \equiv A + Bx + Cx^2 + \ldots$$

find the values of the integers A, B and C. **(4 marks)**

(E) **6** **a** Expand $(1 - 2x)^{10}$ in ascending powers of x up to and including the term in x^3, simplifying each coefficient in the expansion. **(4 marks)**

b Use your expansion to find an approximation of 0.98^{10}, stating clearly the substitution which you have used for x. **(3 marks)**

(E) **7** **a** Use the binomial series to expand $(2 - 3x)^{10}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer. **(4 marks)**

b Use your series expansion, with a suitable value for x, to obtain an estimate for 1.97^{10}, giving your answer to 2 decimal places. **(3 marks)**

(E/P) **8** **a** Expand $(3 + 2x)^4$ in ascending powers of x, giving each coefficient as an integer. **(4 marks)**

b Hence, or otherwise, write down the expansion of $(3 - 2x)^4$ in ascending powers of x. **(2 marks)**

c Hence by choosing a suitable value for x show that $(3 + 2\sqrt{2})^4 + (3 - 2\sqrt{2})^4$ is an integer and state its value. **(2 marks)**

(E/P) **9** The coefficient of x^2 in the binomial expansion of $\left(1 + \frac{x}{2}\right)^n$, where n is a positive integer, is 7.

a Find the value of n. **(2 marks)**

b Using the value of n found in part **a**, find the coefficient of x^4. **(4 marks)**

(E) **10** **a** Use the binomial theorem to expand $(3 + 10x)^4$ giving each coefficient as an integer. **(4 marks)**

b Use your expansion, with an appropriate value for x, to find the exact value of 1003^4. State the value of x which you have used. **(3 marks)**

(E) **11** **a** Expand $(1 + 2x)^{12}$ in ascending powers of x up to and including the term in x^3, simplifying each coefficient. **(4 marks)**

b By substituting a suitable value for x, which must be stated, into your answer to part **a**, calculate an approximate value of 1.02^{12}. **(3 marks)**

c Use your calculator, writing down all the digits in your display, to find a more exact value of 1.02^{12}. **(1 mark)**

d Calculate, to 3 significant figures, the percentage error of the approximation found in part **b**. **(1 mark)**

(E/P) **12** Expand $\left(x - \frac{1}{x}\right)^5$, simplifying the coefficients. **(4 marks)**

(E/P) **13** In the binomial expansion of $(2k + x)^n$, where k is a constant and n is a positive integer, the coefficient of x^2 is equal to the coefficient of x^3.

a Prove that $n = 6k + 2$. **(3 marks)**

b Given also that $k = \frac{2}{3}$, expand $(2k + x)^n$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an exact fraction in its simplest form. **(4 marks)**

E/P **14** **a** Expand $(2 + x)^6$ as a binomial series in ascending powers of x, giving each coefficient as an integer. **(4 marks)**

b By making suitable substitutions for x in your answer to part **a**, show that $(2 + \sqrt{3})^6 - (2 - \sqrt{3})^6$ can be simplified to the form $k\sqrt{3}$, stating the value of the integer k. **(3 marks)**

E/P **15** The coefficient of x^2 in the binomial expansion of $(2 + kx)^8$, where k is a positive constant, is 2800.

a Use algebra to calculate the value of k. **(2 marks)**

b Use your value of k to find the coefficient of x^3 in the expansion. **(4 marks)**

E/P **16** **a** Given that

$$(2 + x)^5 + (2 - x)^5 \equiv A + Bx^2 + Cx^4,$$

find the value of the constants A, B and C. **(4 marks)**

b Using the substitution $y = x^2$ and your answers to part **a**, solve

$$(2 + x)^5 + (2 - x)^5 = 349.$$ **(3 marks)**

E/P **17** In the binomial expansion of $(2 + px)^5$, where p is a constant, the coefficient of x^3 is 135. Calculate:

a the value of p, **(4 marks)**

b the value of the coefficient of x^4 in the expansion. **(2 marks)**

P **18** Find the constant term in the expansion of $\left(\frac{x^2}{2} - \frac{2}{x}\right)^9$.

E/P **19** **a** Find the first three terms, in ascending powers of x of the binomial expansion of $(2 + px)^7$, where p is a constant. **(2 marks)**

The first 3 terms are 128, $2240x$ and qx^2, where q is a constant.

b Find the value of p and the value of q. **(4 marks)**

E/P **20** **a** Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 - px)^{12}$, where p is a non-zero constant. **(2 marks)**

b Given that, in the expansion of $(1 - px)^{12}$, the coefficient of x is q and the coefficient of x^2 is $6q$, find the value of p and the value of q. **(4 marks)**

E/P **21** **a** Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\left(2 + \frac{x}{2}\right)^7$, giving each term in its simplest form. **(4 marks)**

b Explain how you would use your expansion to give an estimate for the value of 2.05^7. **(1 mark)**

E/P **22** $g(x) = (4 + kx)^5$, where k is a constant.

Given that the coefficient of x^3 in the binomial expansion of $g(x)$ is 20, find the value of k. **(3 marks)**

Challenge

1 $f(x) = (2 - px)(3 + x)^5$ where p is a constant.
There is no x^2 term in the expansion of $f(x)$.
Show that $p = \frac{4}{3}$

2 Find the coefficient of x^2 in the expansion of $(1 + 2x)^8(2 - 5x)^7$.

Summary of key points

1 Pascal's triangle is formed by adding adjacent pairs of numbers to find the numbers on the next row.

2 The $(n + 1)$th row of Pascal's triangle gives the coefficients in the expansion of $(a + b)^n$.

3 $n! = n \times (n - 1) \times (n - 2) \times \ldots \times 3 \times 2 \times 1$.

4 You can use factorial notation and your calculator to find entries in Pascal's triangle quickly.

- The number of ways of choosing r items from a group of n items is written as ${}^nC_r$ or $\binom{n}{r}$: ${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$
- The rth entry in the nth row of Pascal's triangle is given by ${}^{n-1}C_{r-1} = \binom{n-1}{r-1}$.

5 The binomial expansion is:

$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n \ (n \in \mathbb{N})$

where $\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$

6 In the expansion of $(a + b)^n$ the general term is given by $\binom{n}{r}a^{n-r}b^r$.

7 The first few terms in the binomial expansion can be used to find an approximate value for a complicated expression.